

Fig. 1. Two-dimensional behavioral classification framework. Priority (vertical) and aggression (horizontal) define four quadrants: Q1 (high priority, high aggression — facilitate), Q2 (high priority, low aggression — protect), Q3 (low priority, low aggression — serve), Q4 (low priority, high aggression — throttle).

Fig. 2. Passive transaction latency and CPU idle rate at four representative aggression levels across four scales. Full 10-level sweep available in V11 simulation notebook. Optimal zone 75-90% produces 15-26% improvement over orderly baseline.

Fig. 3. Passive transaction latency comparison across three assignment methods at 85% aggression. Method C (Convergent Random) achieves 60% improvement over the orderly baseline at 5000 agents with 4x lower variance.

Fig. 4. Complete performance comparison at 5,000 agents. Each bar represents passive transaction latency — shorter is better. Method C achieves 60% reduction over orderly baseline.

Fig. 5. Redis SET throughput and P50 latency across concurrency levels $c=1$ to $c=1000$ (three-run validated, 200,000 ops/run). Throughput plateau at $c=30-100$ confirms BPQT optimal zone. Latency spike above $c=200$ confirms thrashing signal.

Fig. 6. BPQT simulation vs Redis benchmark results, normalized to passive baseline = 1.0x. Both confirm consistent improvement trajectory across behavioral scenarios.

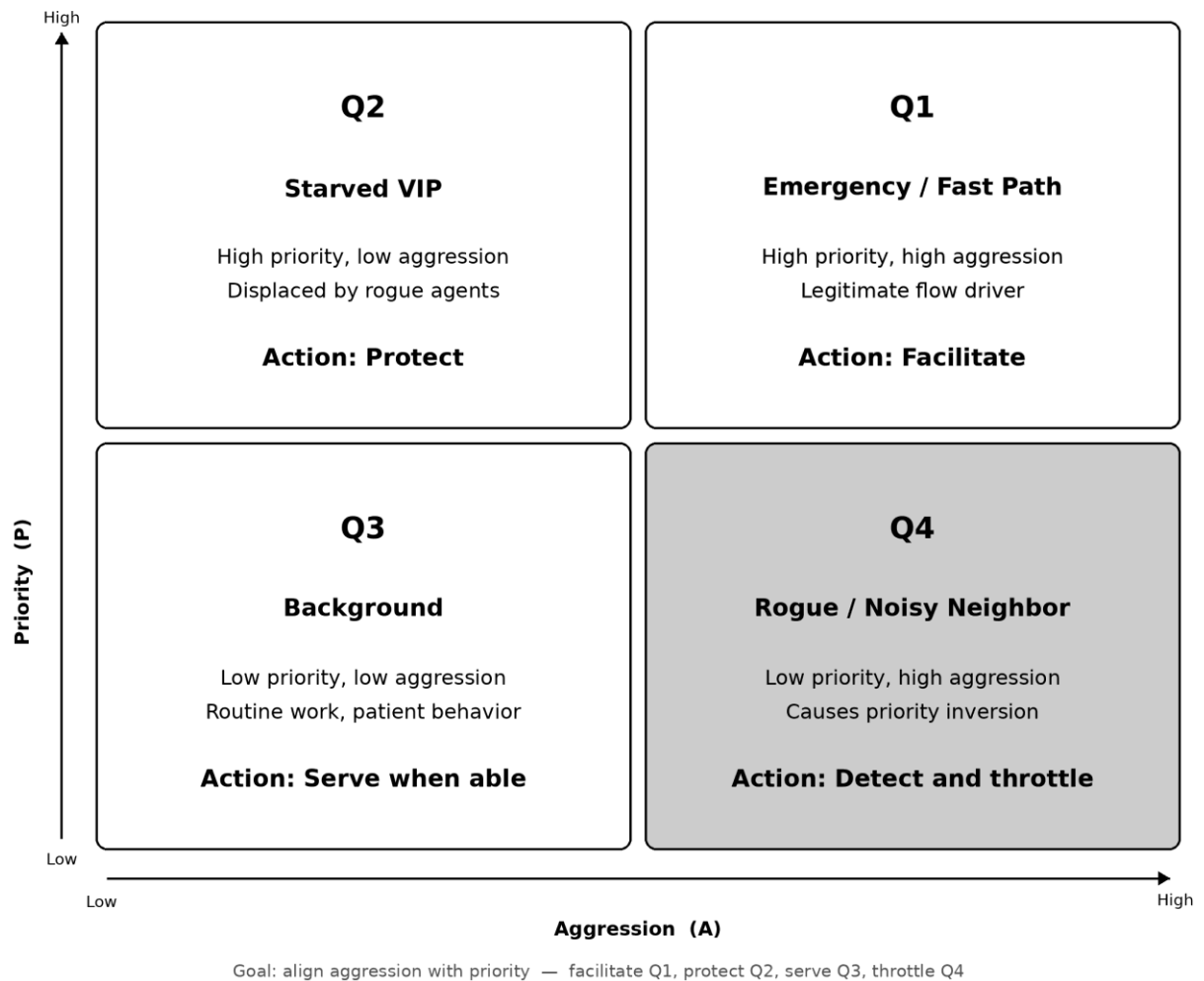


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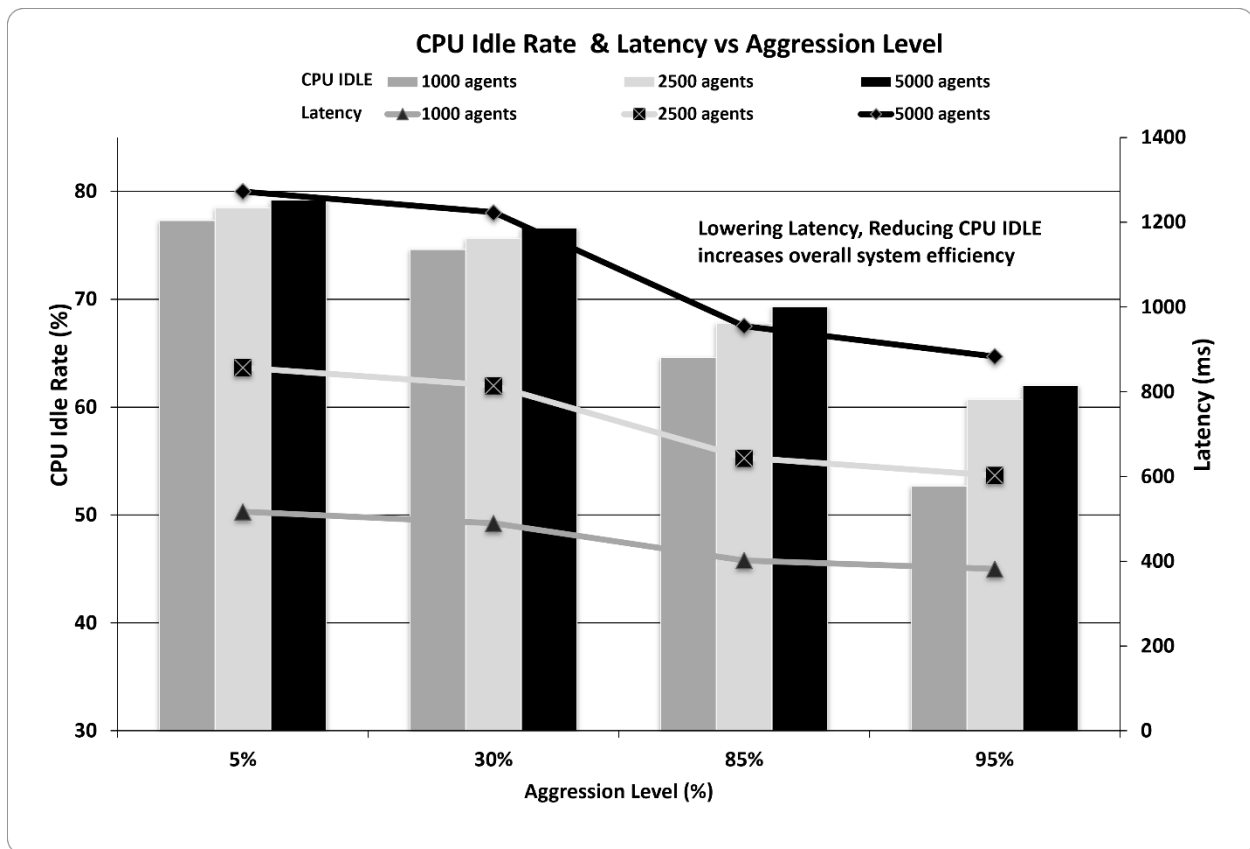


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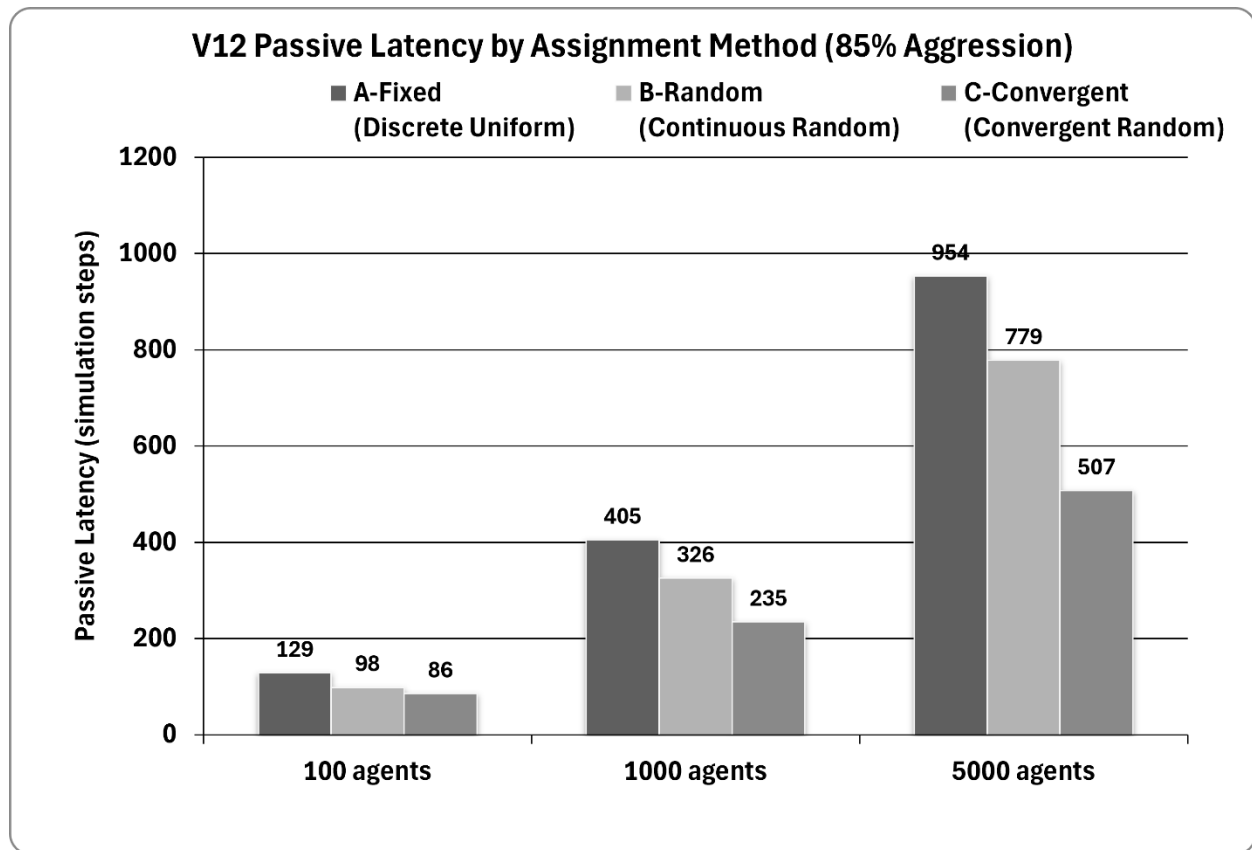


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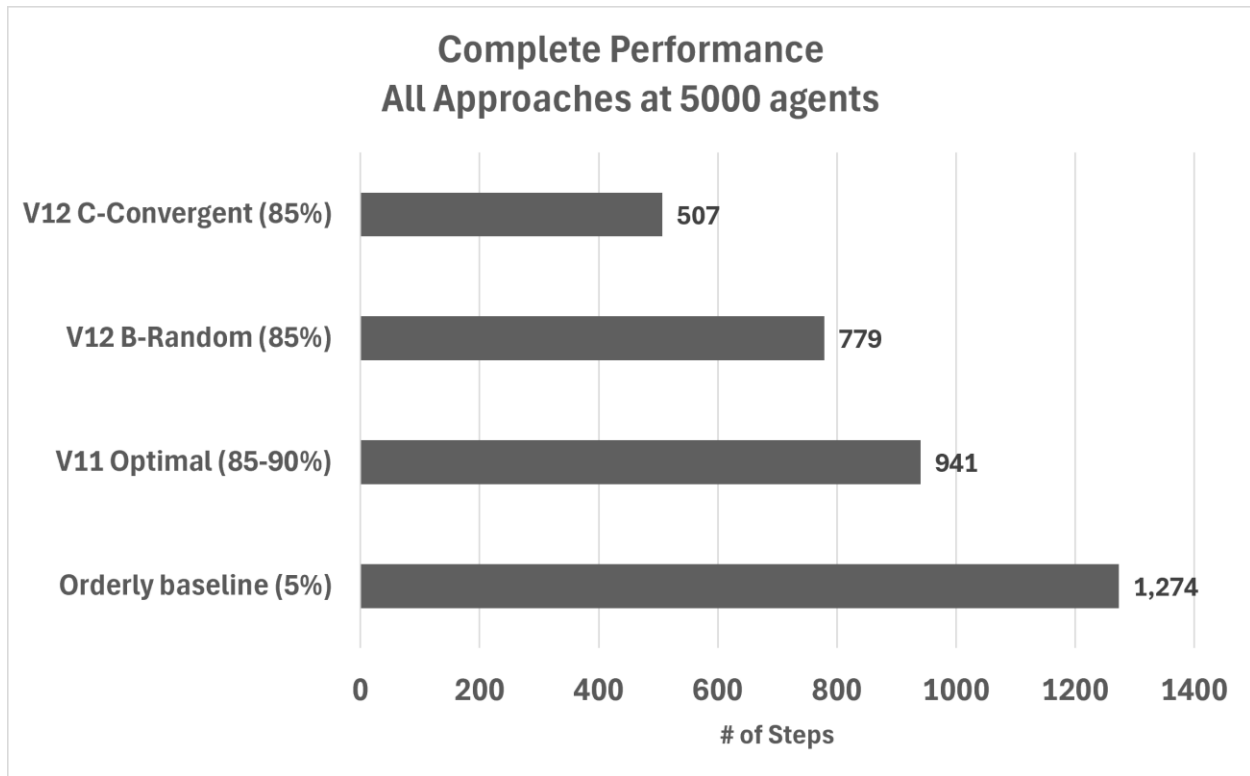


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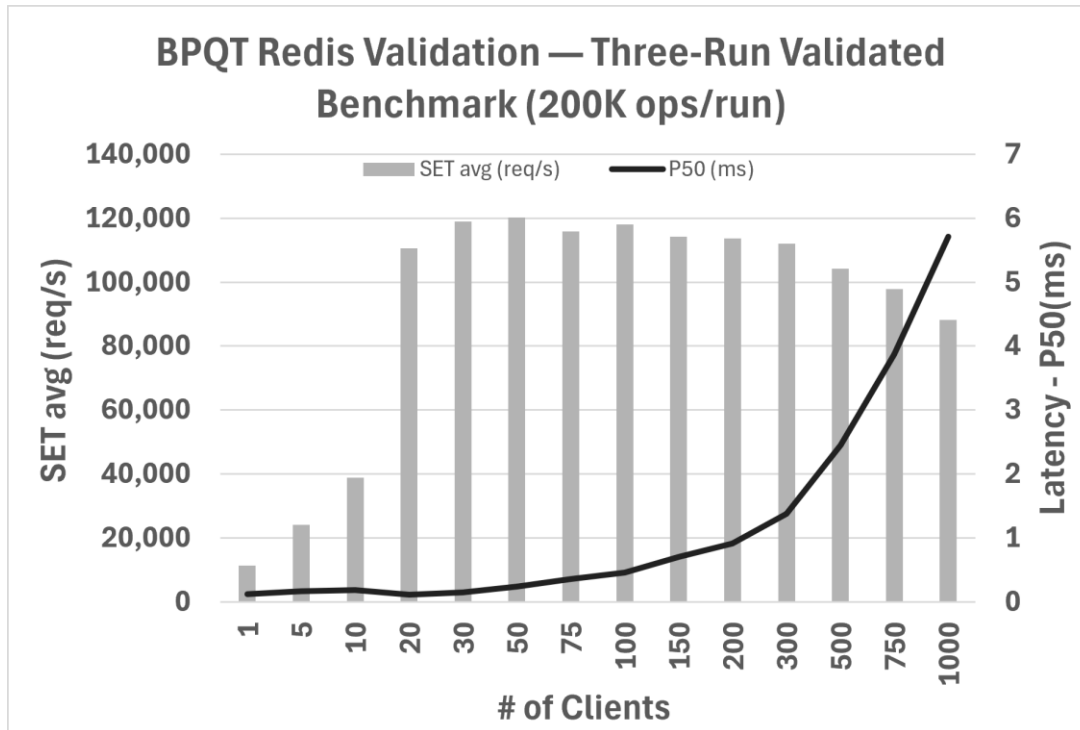


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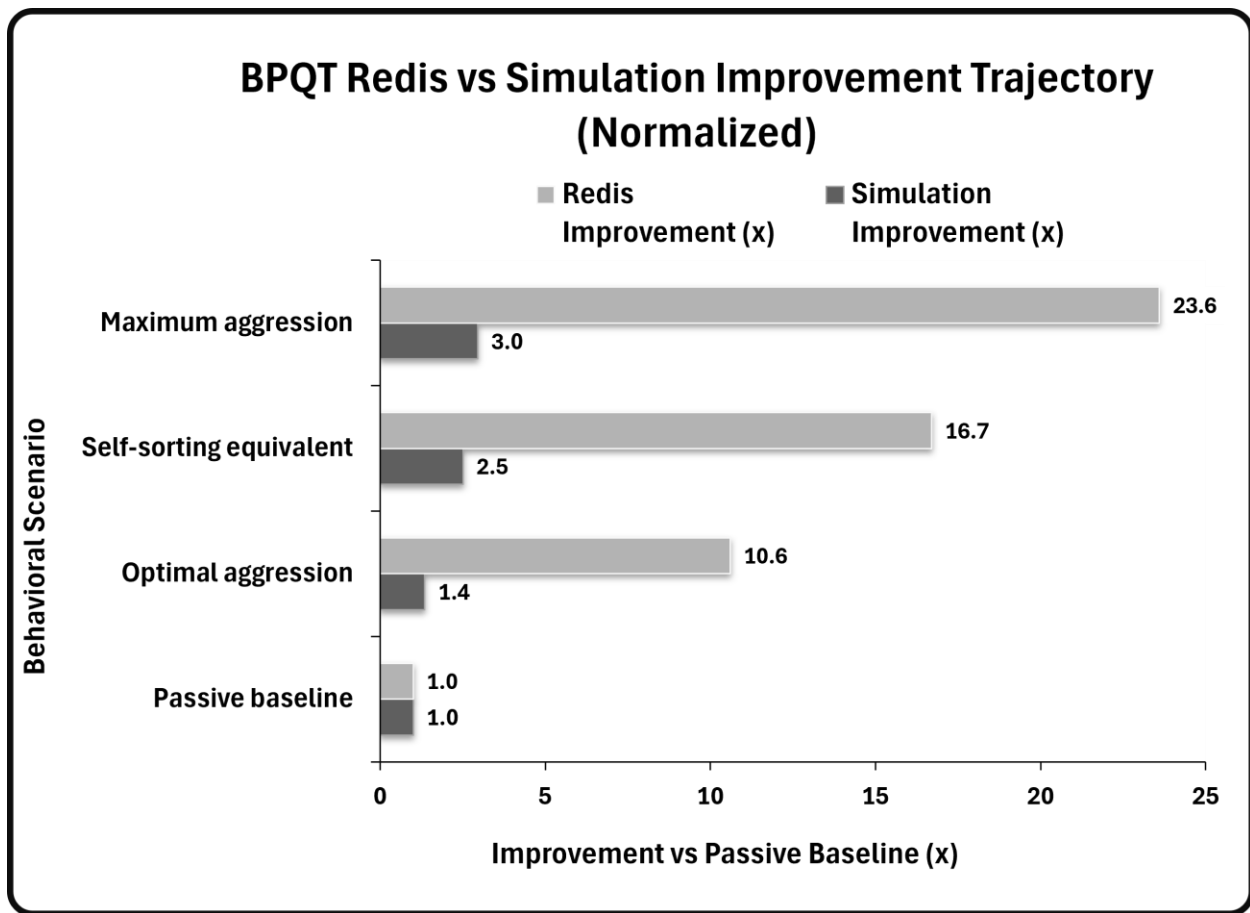


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